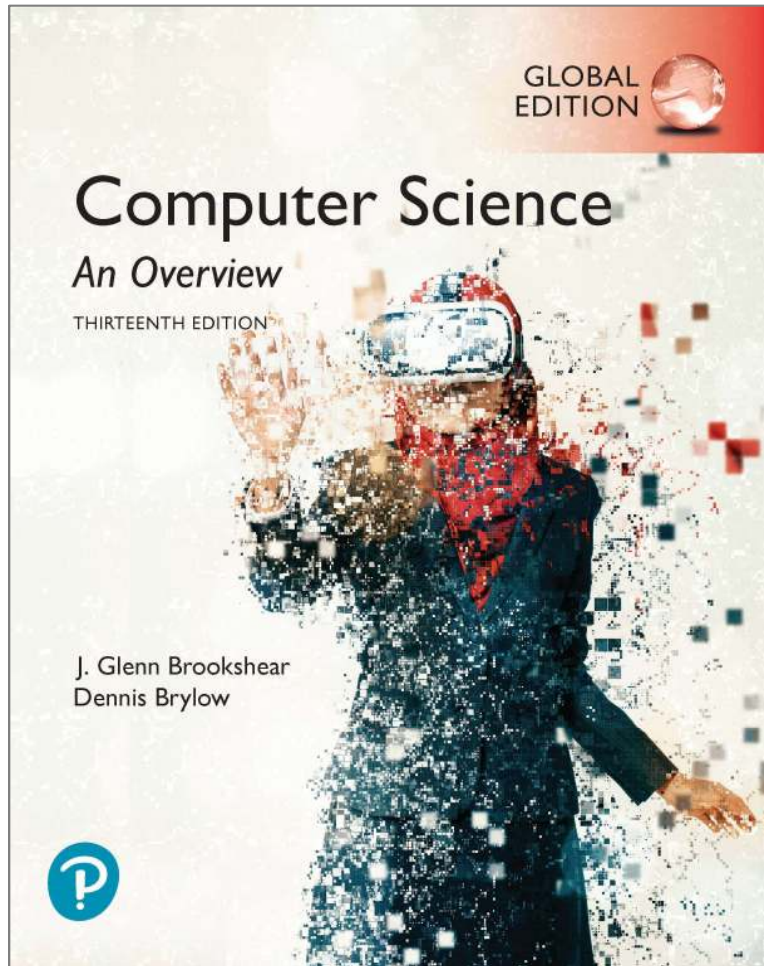


Computer Science An Overview

13th Edition, Global Edition



Chapter 4

Networking and the Internet

4.1 Network Fundamentals

- The need to share information and resources among different computers has led to linked computer systems, called **networks**, in which computers are connected so that data can be transferred from machine to machine.
- Network Software allows users to exchange information and share resources
 - Content
 - Software
 - Data storage facilities

Network Classifications

- Scope
 - Personal Area Network (short-range)
 - Local Area Network (building/campus)
 - Metropolitan Area Network (community)
 - Wide Area Network (greater distances)
- Topology (configuration)
 - Bus (Ethernet)
 - Star (Wireless networks with central Access Point)

Figure 4.1 Two popular network topologies (1 of 2)

a. Bus

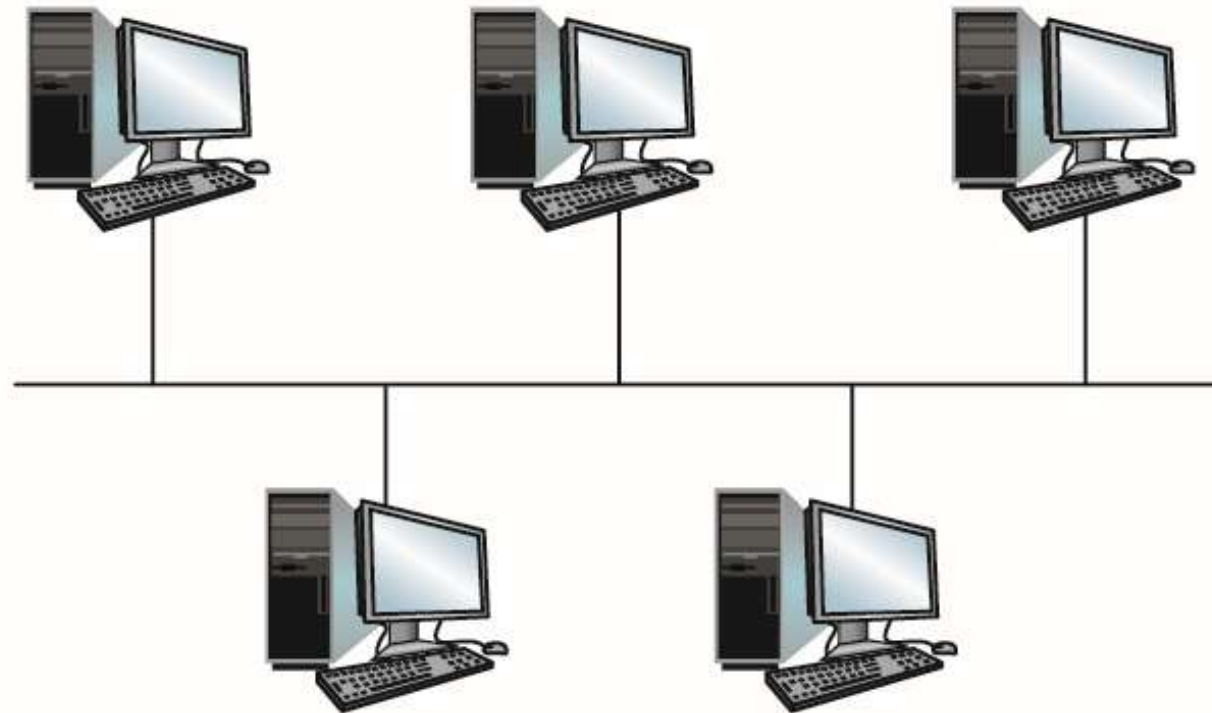
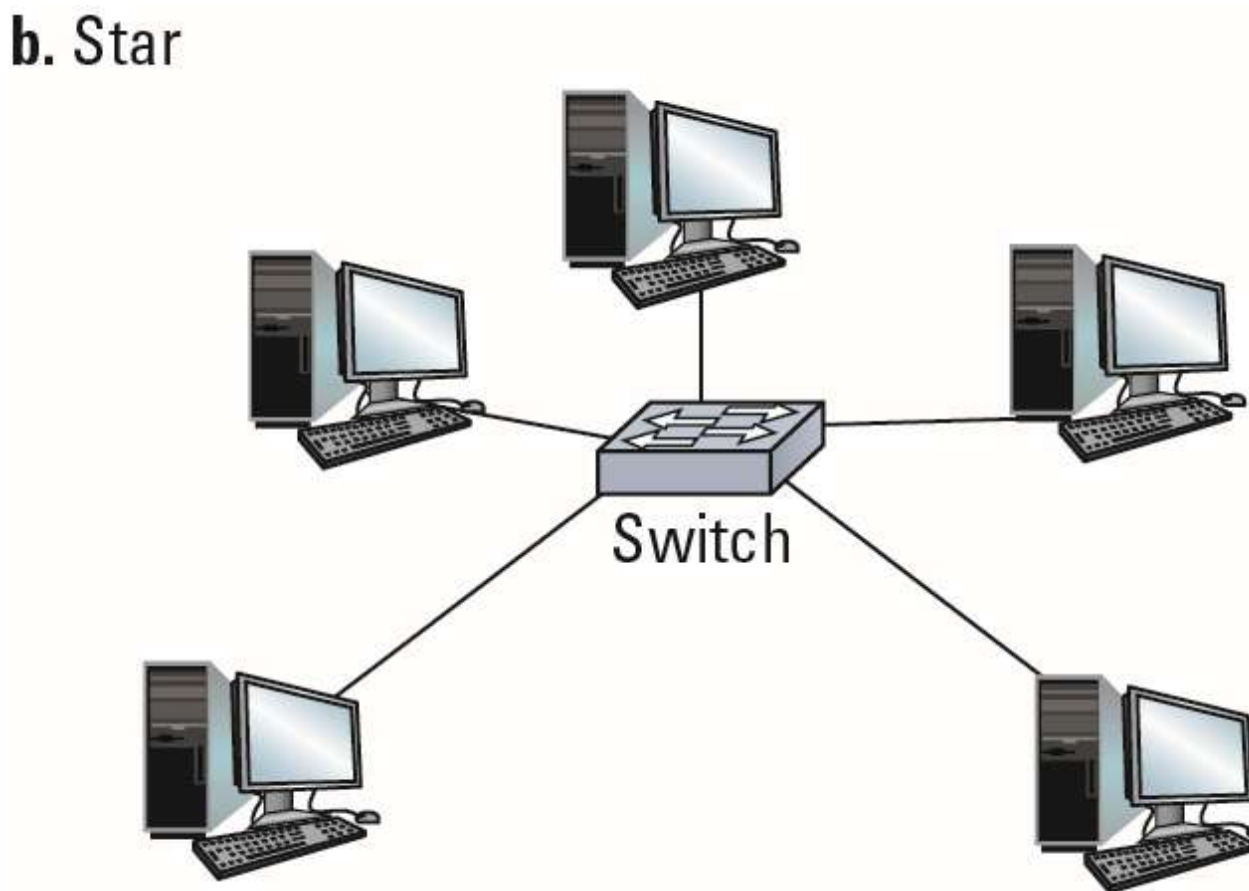


Figure 4.1 Two popular network topologies (2 of 2)

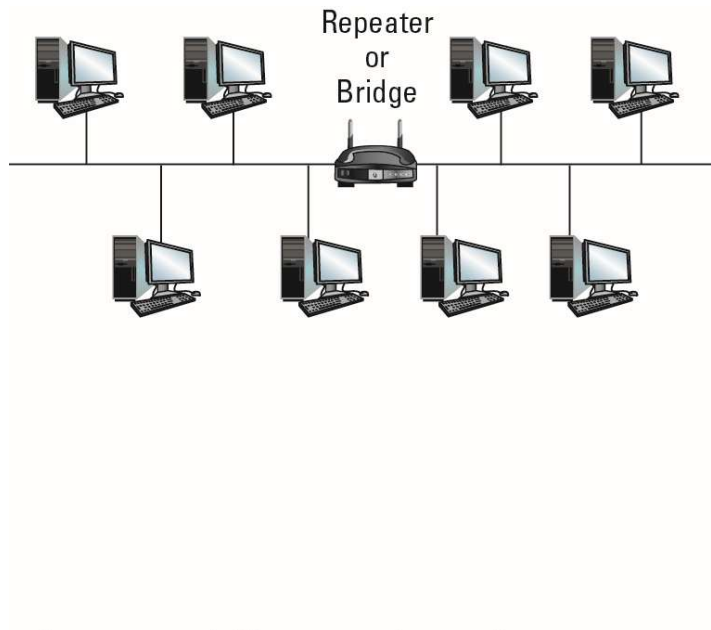
b. Star



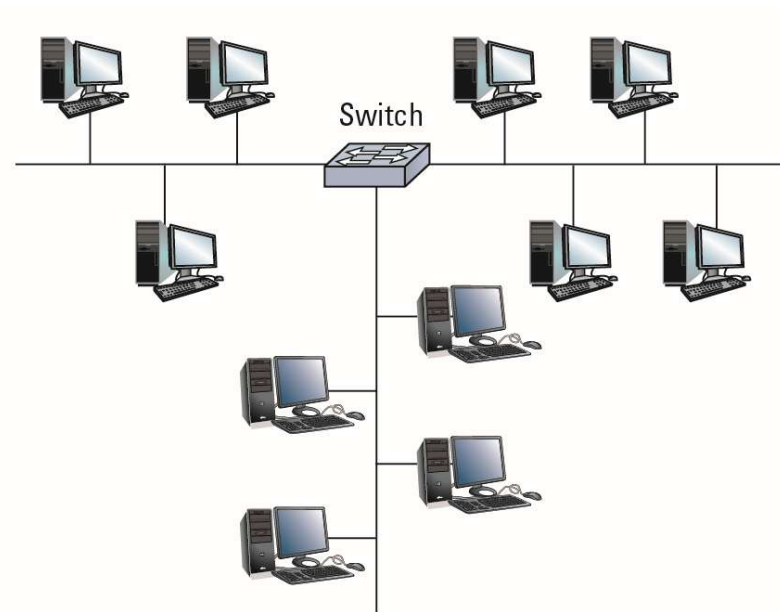
Combining Networks

- **Repeater:** passes all messages across two busses
- **Bridge:** passes only messages that are destined for computers on the other bus
- **Switch:** acts like a bridge, but with connections to multiple busses
- **Router:** Connects two incompatible networks resulting in a network of networks called an **internet**

Figure 4.4 Building a large bus network from smaller ones

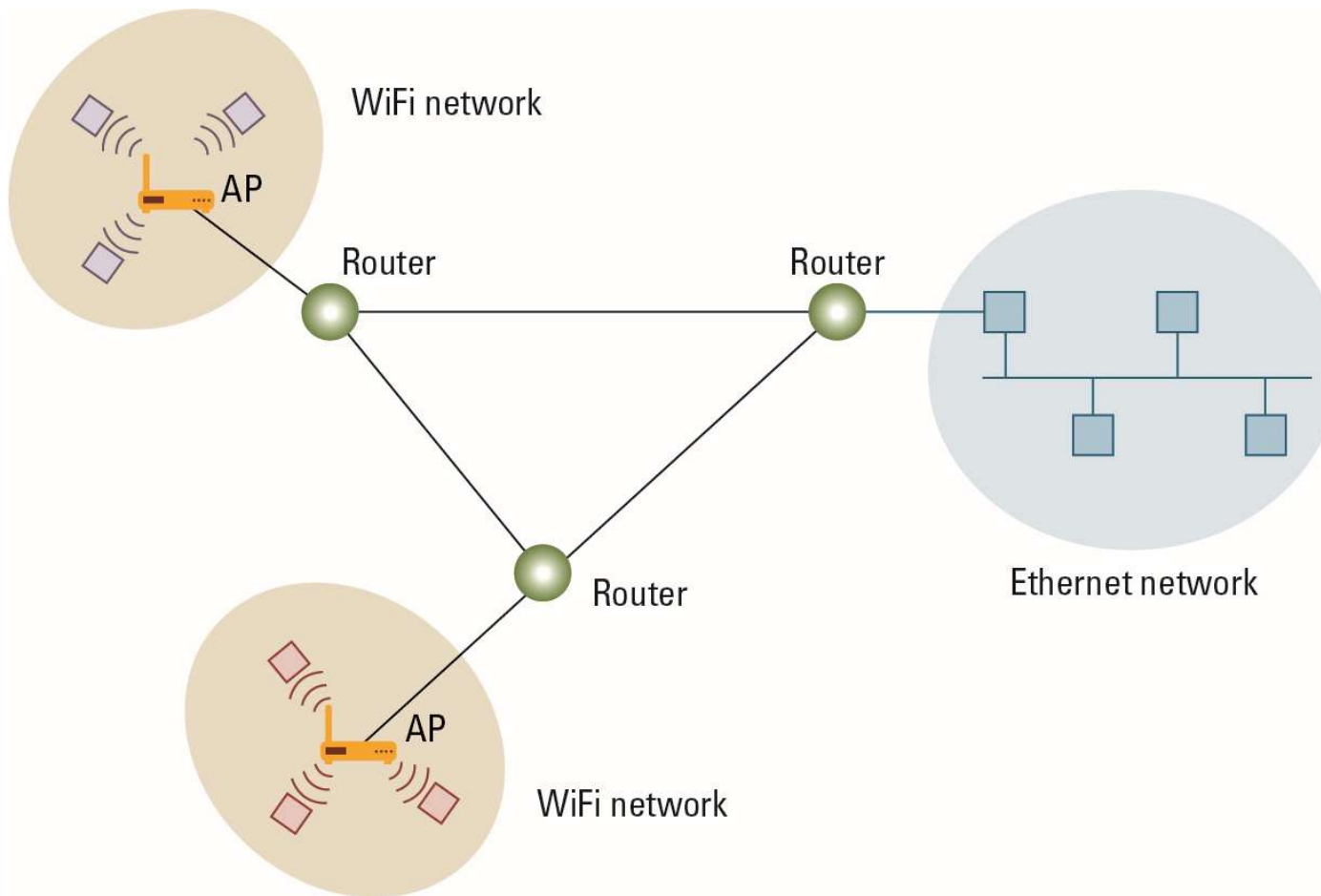


a. A repeater or bridge connecting two buses



b. A switch connecting multiple buses

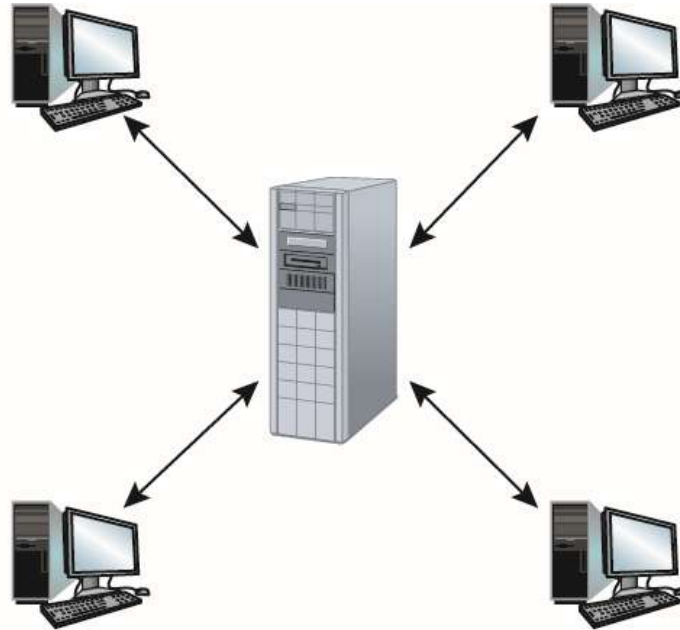
Figure 4.5 Routers connecting two WiFi networks and an Ethernet network to form an internet



Methods of Process Communication

- Client-server
 - Many clients, one server (executing continuously)
 - Clients make requests of other processes
 - Server satisfies requests made by clients
- Peer-to-peer (P2P)
 - Two processes communicating as equals
 - Processes execute on a temporary basis

Figure 4.6 The client/server model compared to the peer-to-peer model



a. Server must be prepared to serve multiple clients at any time.



b. Peers communicate as equals on a one-to-one basis.

Distributed Systems

- Systems units that execute as processes on different computers
 - Cluster computing
 - Independent computers work closely together instead of a single, much larger machine
 - Grid computing
 - Millions of home PCs (not connected to each other) work on a complex problem
 - Cloud computing
 - Provide services, hide the details

4.2 The Internet

- The Internet is an internet that spans the world
 - Original goal was to link a variety of networks into a connected system unaffected by local disasters
 - Today, it is a commercial undertaking that links a worldwide combination of PANs, LANs, MANs, and WANs involving millions of computers

Internet Addressing

- IP address: pattern of 32 or 128 bits often represented in dotted decimal notation
- Mnemonic address:
 - Domain names (mu.edu)
 - Top-Level Domains
 - .org, .gov, .com, .mil, .net, .au, .ca, .biz,
- Domain name system (DNS)
 - Name servers
 - DNS lookup

Early Internet Applications

- Network News Transfer Protocol (NNTP)
- File Transfer Protocol (FTP)
- Telnet and Secure Shell (SSH)
- Hypertext Transfer Protocol (HTTP)
- Electronic Mail (email)
 - Domain mail server collects incoming mail and transmits outgoing mail
 - Mail server delivers collected incoming mail to clients via POP3 or IMAP

More Recent Applications

- Voice Over IP (VoIP)
- Internet Multimedia Streaming
 - N-unicast
 - Multicast
 - On-demand streaming
 - Content delivery networks (CDNs)